

Name: _____

Solutions

Date: _____

SAFETY BAY
SENIOR HIGH SCHOOL*imagine believe achieve*

Year 10 Mathematics

Investigation 2, 2015

Topic – Sketching Quadratics

Validation Section

Total Time:

50 minutes

Weighting:

10% of the year

Equipment:

Sketching Quadratics Take-Home Section

Scientific Calculators are allowed.**No CAS or Graphing Calculators to be used.**

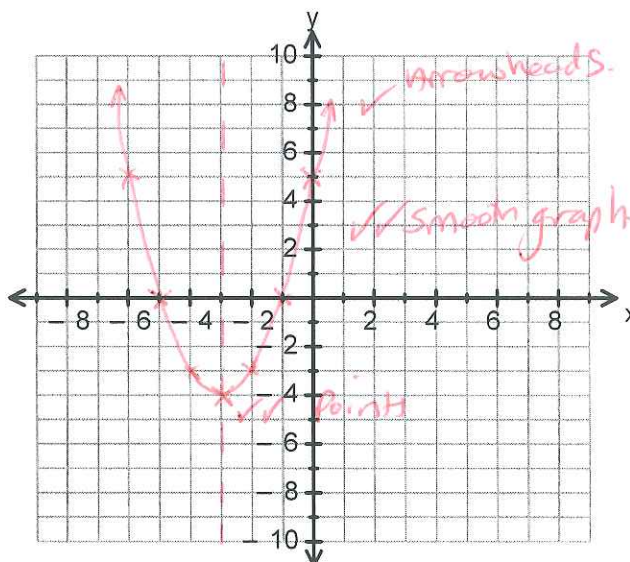
70 %

Marks given for x and y intercepts, axis of symmetry, turning point, plotting correct points, sketching a smooth graph, putting arrow heads on the ends of the lines. Each question is worth 10 marks

Sketch the following parabolas and list their key features.

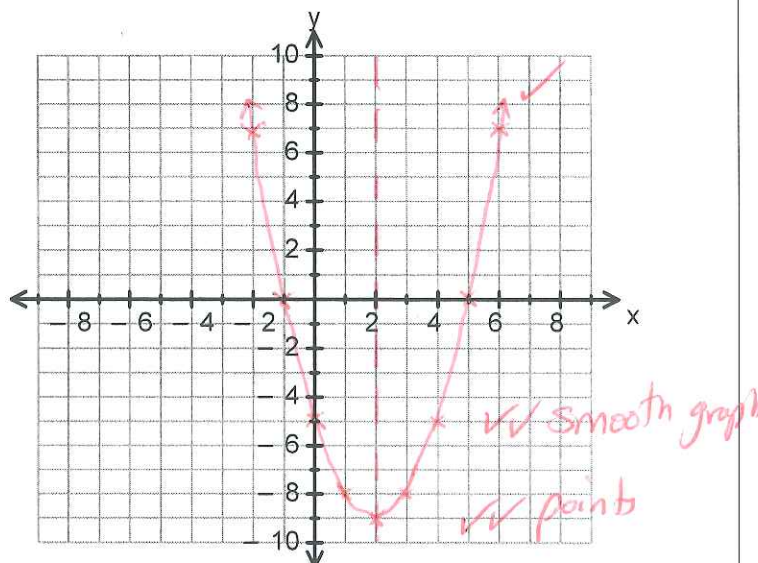
1. $y = (x + 5)(x + 1)$

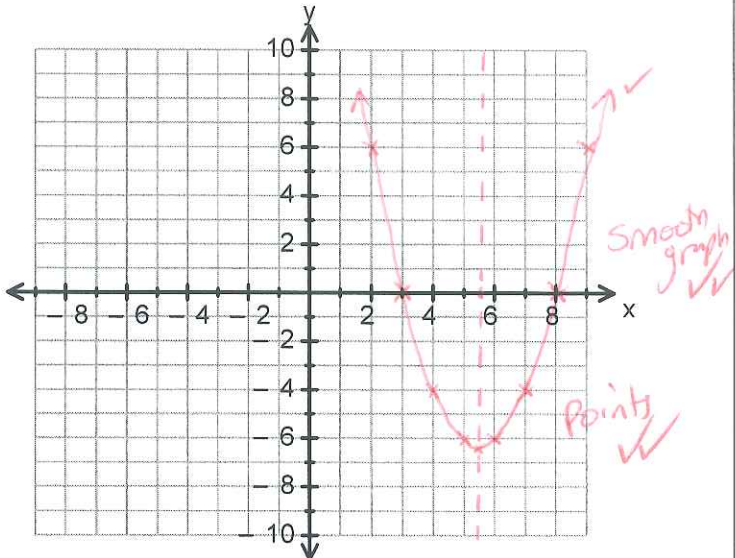
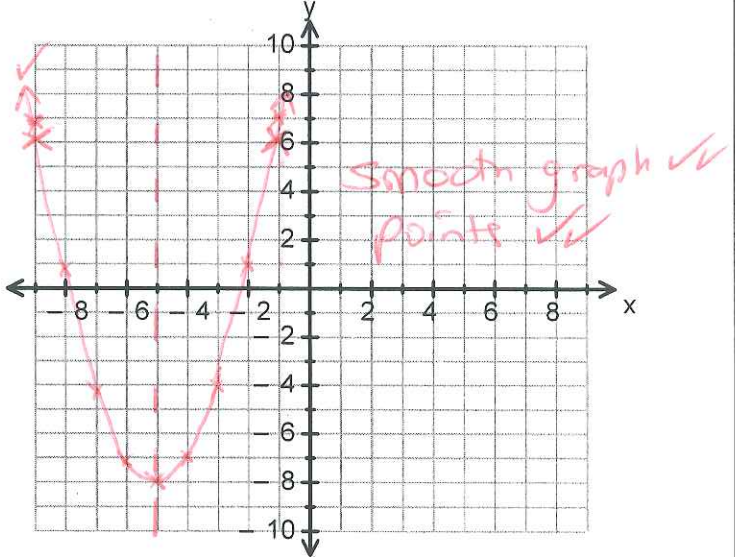
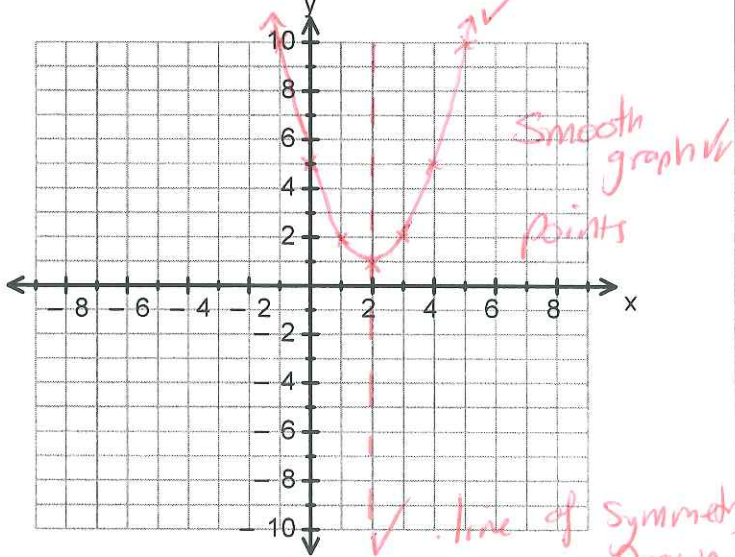
y-intercept (0, 5) ✓
x-intercepts (-5, 0) and (-1, 0) ✓
axis of symmetry: $x = -3$ ✓
Turning Point (-3, -4) ✓

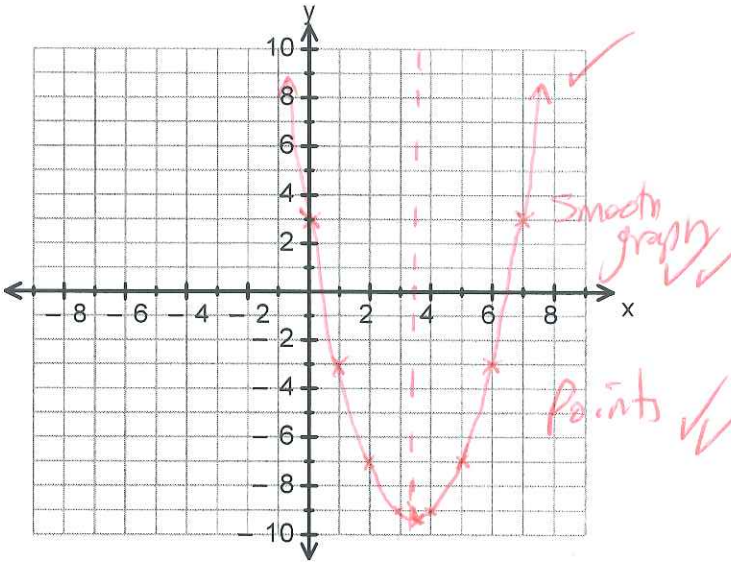
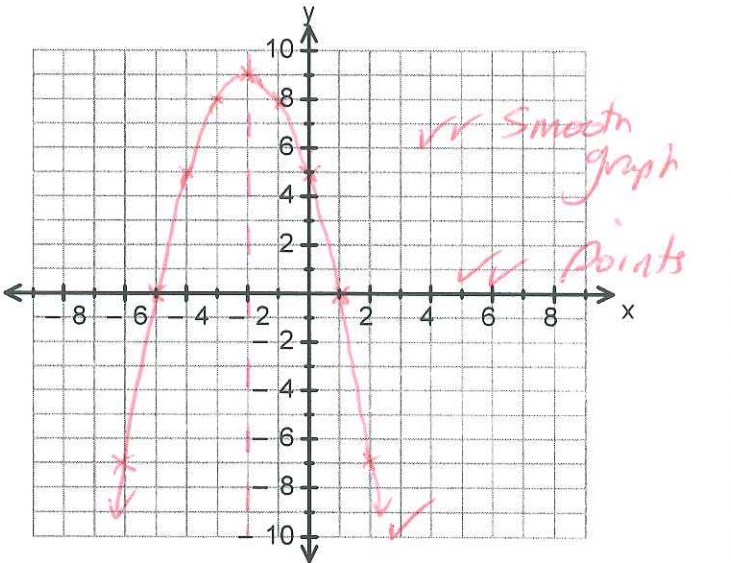


2. $y = (x - 5)(x + 1)$

y-intercept (0, -5) ✓
x-intercepts (5, 0) and (-1, 0) ✓
axis of symmetry: $x = 2$ ✓
Turning Point (2, -9) ✓



3. $y = (x - 8)(x - 3)$ <i>not on graph</i> ✓ y-intercept $(0, 24)$ ✓ x-intercepts $(8, 0)$ & $(3, 0)$ ✓ ✓ Axis of symmetry: $x = 5\frac{1}{2}$ ✓ Turning Point $(5\frac{1}{2}, -6.25)$ ✓ $(3, -6), (4, -4), (2, 6)$	
4. $(x + 5)^2 - 8$ y-intercept <i>not on graph</i> or $(0, 13)$ ✓ x-intercepts <i>Approximate</i> $(-7.8, 0)$ & $(-2.2, 0)$ ✓ ✓ Axis of symmetry: $x = -5$ ✓ Turning Point $(-5, -8)$ ✓	
5. $(x - 2)^2 + 1$ y-intercept $(0, 5)$ ✓ x-intercepts none ✓ Axis of symmetry: $x = 2$ ✓ Turning Point $(2, 1)$ ✓	

6.	$y = x^2 - 7x + 3$ y-intercept $(0, 3)$ ✓ x-intercepts $(0.5, 0)$ & $(6.5, 0)$ ✓ approximate line of symmetry: $x = 3\frac{1}{2}$ ✓ Turning Point $(3\frac{1}{2}, -9.25)$ ✓ $(3, -9), (2, -7), (1, -3)$	
7.	$y = -x^2 - 4x + 5$ y-intercept $(0, 5)$ ✓ x-intercepts $(-5, 0)$ & $(1, 0)$ ✓ Axis of Symmetry: $x = -2$ ✓ Turning Point $(-2, 9)$ ✓	

End of Validation